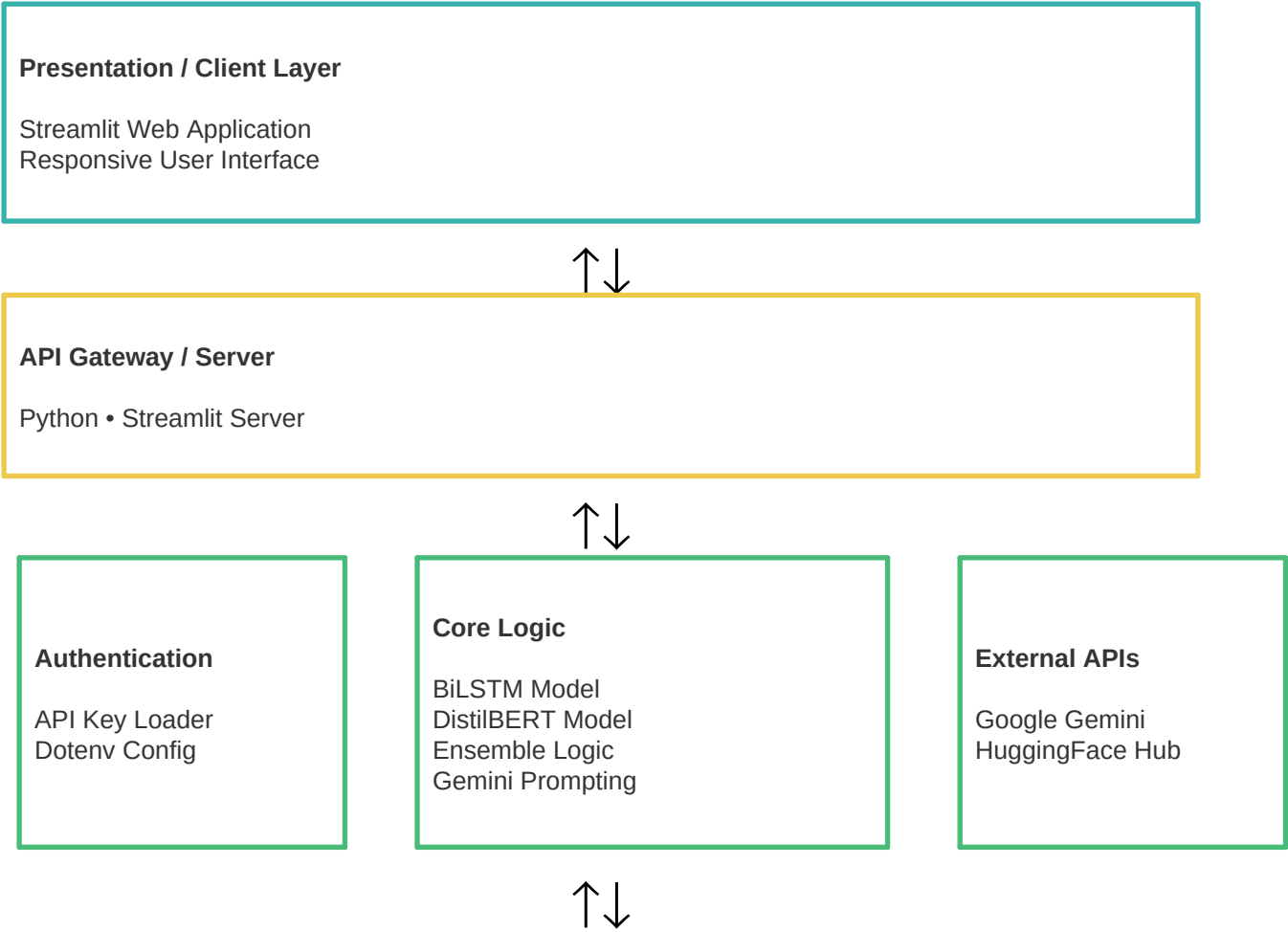


Solution Architecture

Date	10 july 2026
Team ID	XXXXXX
Project Name	Emotion-Aware Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram

The Emotion-Aware Learning Support Engine follows a layered architecture consisting of Presentation, API, Core Logic, and Data layers. Text is uploaded through the client interface, processed by ensemble AI models, and stored securely along with interactions.



Data / Storage Layer

CSV Logs • Model Weights (.npz/.safetensors)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for text input, result viewing and dashboard interaction.	HTML, CSS, Streamlit, Plotly
API Gateway	Routes requests between the client interface and AI services.	Python, Streamlit
Core Logic Service	Performs preprocessing, ensemble emotion classification, and Gemini guidance generation.	TensorFlow, PyTorch, Google GenAI
Database / Storage	Stores interaction logs, model weights, and tokenizer files locally.	CSV, NumPy, JSON